TARUNKUMAR B T

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Bangalore, India

Experience

AI Engineer (Conversational AI Engineer) Feb 2026 – Present
SoundHound AI *Bangalore, India*

- **Compass | Real-Time Agentic Voice AI:** Developed a production-grade healthcare voice agent using **OpenAI Realtime, multi-agent orchestration, and tool calling**, enabling natural-language search across **400+ menu items**, dynamic item customization, and stateful multi-turn meal-ordering workflows integrated with enterprise APIs.
- Implemented patient-aware ordering guardrails for **allergies, dietary restrictions, nutrients, meal availability, quantity limits**, and stateful tray updates; supported delivery status and order cancellation.
- Improved real-time voice interactions with contextual recommendations, English/Spanish switching, natural speech responses, and safeguards against unintended tool calls or tray changes.
- **Acentra | IT Service Desk:** Orchestrated specialized AI agents for identity verification, Duo MFA, password resets, account unlocks, application-specific routing, case management, onboarding, and human escalation.
- **PCL | IT Help Desk:** Developed knowledge-base-grounded troubleshooting with verified article retrieval, step-by-step assistance, knowledge article delivery, ticket creation, and escalation workflows.
- **MCP Developer Tooling:** Built a FastMCP-based Amelia integration/testing tool adopted by **approximately 80% of the team** for multi-turn tests, conversation-history review, workflow validation, and escalation tracking.
- **Audio Automation:** Developed Python pipelines to bulk-download call recordings from CSV/Excel inputs and support audio classification and call-quality analysis in an internal dashboard.

AI/ML Engineer Feb 2025 – Sep 2025
Redwoods.ai *Hyderabad, India*

- Designed and deployed a production **RAG search system** using LlamaIndex, embeddings, and Qdrant, improving retrieval precision by **35%** compared with keyword-based search.
- Built asynchronous document ingestion, chunking, and embedding pipelines for large enterprise files; optimized retrieval context and token utilization for LLM queries.
- Built **LangGraph agents** integrating web search, retrieval, and backend actions; automated supplier onboarding with ESG report analysis, FAQ generation, and website data extraction.
- Optimized request handling with batching and parallel processing, reducing response latency by **40%**; contributed to model evaluation and production delivery.

Python Developer Jul 2024 – Dec 2024
ITMTB Technologies Pvt. Ltd. *Pune, India*

- Developed Python-based data migration and synchronization pipelines to integrate Shopify and Myntra product data through Increff and Clickpost APIs.
- Implemented LLM-assisted product creation and SEO-content generation workflows, reducing manual effort by **80%**.

Personal Projects

BotLattice | Full-Stack RAG SDK GitHub

- Architected a configurable **end-to-end RAG platform** with FastAPI and React for document ingestion, chunking, embeddings, vector-store integration, and retrieval workflows.
- Implemented PostgreSQL state management, authentication, **WebSocket streaming**, and Docker-based deployment.

Real-Time AI Voice Tutor | LiveKit + LangGraph

- Built an interactive AI voice tutor with **LiveKit, OpenAI Realtime, LangGraph, and RAG retrieval** for personalized, context-aware multi-turn learning.
- Optimized asynchronous audio processing, barge-in handling, and retrieval-backed replies for conversational continuity.

SkillSync | AI Resume Matching Agent Live Demo | GitHub

- Developed a resume-to-job matching application using **LlamaIndex and OpenAI embeddings** to compare candidate profiles with job descriptions and estimate fit scores.
- Generated skill-gap insights and role-specific suggestions to help users tailor their resumes.

Technical Skills

Programming: Python, Groovy, SQL, Pandas, NumPy, Asyncio
Generative AI: LLMs, RAG, Agentic AI, Multi-Agent Orchestration, Function Calling, Prompt Engineering, LLM Evaluation, Guardrails, Embeddings, MCP
AI Frameworks: LangChain, LangGraph, LlamaIndex, LangSmith, FastMCP, PyTorch, Scikit-learn
Voice AI: LiveKit, OpenAI Realtime, Speech-to-Text (STT), Text-to-Speech (TTS), Conversational AI
Backend: FastAPI, Flask, REST APIs, Async Programming,PostgreSQL, MySQL, Qdrant (Vector DB)
Cloud & DevOps: Docker, AWS S3, Azure Blob Storage, Git

Education & Certifications

B.E. Computer Science and Engineering, Dr. AIT, Bangalore 2020 – 2024 | CGPA: 8.33
Certifications: LangGraph (IBM) | Machine Learning (Coursera) | LangChain (DeepLearning.AI) | Advanced Learning Algorithms (Stanford)